

Supporting information

S2 Algorithm optimisation

A pairwise comparison of trait values costs more computation time than a single

- 5 comparison to the mean. As expected, the simulations for the pairwise competition
model were slower (around 4 times) than for the mean competition model. Yet, the
computation time is a critical issue in a context of large numbers of simulation
experiments required for classifier training, and this necessitated code optimization.

<u>Model</u>	<u>Before optimization:</u> <u>[29] with pairwise</u> <u>competition added by</u> <u>this study</u>	<u>After</u> <u>optimization:</u> <u>this study</u>	<u>Ratio</u>
<u>Neutral</u>	<u>9 ± 2</u>	<u>24 ± 3</u>	<u>0.375</u>
<u>Mean competition</u>	<u>456 ± 114</u>	<u>89 ± 11</u>	<u>5.14</u>
<u>Pairwise competition</u>	<u>1944 ± 258</u>	<u>323 ± 40</u>	<u>6.02</u>
<u>Environmental filtering</u>	<u>80 ± 18</u>	<u>70 ± 10</u>	<u>1.14</u>

S2 Table 2: Comparison of the speed of the simulations for different
version of the code (mean value per run, ± standard deviation, in seconds).

Results are for 50 simulations of 200 generations on a single core. According

to the 2018 census Beckenham, a suburb south of Christchurch's central city has the highest commuter cycling rates in New Zealand. Why? And what can be learned from Beckenham that could help other suburbs increase their cycling rates? This research used a range of methods including analysis of census data, an analysis of the local geography, a community survey and focus groups of local residents, to identify factors that may encourage people to cycle. Research shows that there is likely a range of factors that influence residents in Beckenham to cycle. These include: perceived safety, filtered permeability facilitated by the presence of the river, community participation, distance to work, workplaces that support cycling, street design and layout, cycle infrastructure and environmental/Personal values. It is concluded that some key initiatives could help encourage cycling in other areas including improving cycling infrastructure, creating filtered permeability, lower speed limits and fostering a sense of cycle community.